

6a	A progressive wave is a wave in which energy is carried from one point to another by means of vibrations or oscillations within the wave.			E	B1												
	A transverse wave is a wave in which the oscillations of the particles in the wave are at right angles to the direction of transfer of energy of the wave.				B1												
6b	Speed, v is defined as distance travelled divided by the time taken. From the definition of wavelength, λ , in one cycle of the source, the wave energy moves a distance λ . The time taken for one cycle is the time period T . Since $f = 1 / T$, wave speed, $v = (\text{ total distance / time taken })$ $v = \frac{\lambda}{T} = f\lambda$			E	B1 B1 A0												
6ci	<table><tr><td>Angle θ</td><td>amplitude</td><td>intensity</td></tr><tr><td>180°</td><td>A</td><td>I</td></tr><tr><td>90°</td><td>0</td><td>0</td></tr><tr><td>60°</td><td>$0.50A$</td><td>$0.25I$</td></tr></table> intensity $\propto \cos^2 \theta$	Angle θ	amplitude	intensity	180°	A	I	90°	0	0	60°	$0.50A$	$0.25I$	A			B1 B1 B1
Angle θ	amplitude	intensity															
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6cii	<table><tr><td>intensity</td><td>angle θ</td></tr><tr><td>zero</td><td>90°</td></tr><tr><td>maximum</td><td>$0^\circ, 180^\circ$</td></tr><tr><td>$\frac{I}{2}$</td><td>$32.8^\circ, 147^\circ$</td></tr></table> Intensity after passing through polaroid Q, $I_Q = I \cos^2 \theta$ Intensity after passing through polaroid R $I_R = I_Q \cos^2 \theta$ $= I \cos^4 \theta$	intensity	angle θ	zero	90°	maximum	$0^\circ, 180^\circ$	$\frac{I}{2}$	$32.8^\circ, 147^\circ$	D			B1 B1				
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7a	When atoms are excited, the electrons move from a lower energy level to a higher			A	M1												